BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: 6191

Retyped faithfully from the original board paper — same questions, no changes.

Mathematics - Paper: I - Objective (i)

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 6191

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: SECTION-A

Q1. Value of (-1)^(-2i/2) is.

(A) -1
(B) 1
(C) i
(D) -i

Q2. The number of elements in power set of {0,1} are.

(A) 4
(B) 3
(C) 2
(D) 1

Q3. If A = [a b; c d] then |A| =

(A) ad+bc
(B) bc-ad
(C) ad-bc
(D) ac-bd

Q4. A square matrix A=[a_ij] is called hermitian matrix if:

(A) $A^t=A$
(B) $A^t=-A$
(C) $(\bar{A})^t=A$
(D) $(\bar{A})^t=-A$

Q5. Which is an exponential equation?

(A) $x^2+1=0$
(B) $x^3+1=0$
(C) $2x+1=0$
(D) $2^x-1=0$

Q6. $\omega^{38} + \omega^{29} + 1 =$

(A) -1
(B) 0
(C) 1
(D) 2

Q7. The rational fraction $(x^2+1)/x^3$ is.

(A) Identity
(B) Irrational
(C) Proper
(D) Improper

Q8. Formula for sum of an infinite geometric series is

(A) $a_1+(n-1)d$
(B) $a_1/(1-r)$
(C) $a_1(1-r^n)/(1-r)$
(D) $a_1(r^n-1)/(r-1)$

Q9. Next two terms of sequence 7, 9, 12, 16, ... are:

(A) 21,27
(B) 21,26
(C) 20,27
(D) 20,26

Q10. Value of 10!/7! is:

(A) 718
(B) 719
(C) 720
(D) 730

Q11. $n(n-1)(n-2)...(n-r+1) =$

(A) n!/r!
(B) n!/(n-r)!
(C) n!/(n+r)!
(D) ${}^nC_r$

Q12. The expansion $1-x+x^2-x^3+... + x^r +... =$

(A) $(1+x)^{-1}$
(B) $(1-x)^{-1}$
(C) $(1+x)^{-2}$
(D) $(1+x)^{-3}$

Q13. The statement $n! > 2^n - 1$ is true for

(A) n=1
(B) n=2
(C) n<4
(D) n≥4

Q14. $1+\cot^2\theta =$

(A) $\cos^2\theta$
(B) $\sin^2\theta$
(C) $\text{cosec}^2\theta$
(D) $\sec^2\theta$

Q15. $\cos^2\alpha - \sin^2\alpha =$

(A) 1
(B) cos 2α
(C) sin 2α
(D) sin 3α

Q16. The period of cosec θ is.

(A) 2π (B) π

(C) π/2 (D) π/6

Q17. a/(2 sin α) =

(A) r (B) r_1

(C) Δ (D) R

Q18. r_1 =

(A) s/(s-a) (B) Δ/(s-a)

(C) Δs/(s-a) (D) Δ/(s+a)

Q19. $\cos(\sin^{-1}(1/\sqrt{2}))$ =

(A) $1/\sqrt{2}$ (B) $-1/\sqrt{2}$

(C) 1/2 (D) -1/2

Q20. Solution of cot θ = $1/\sqrt{3}$ which lies in [0, 2π]

(A) π/3 (B) π/4

(C) π/6 (D) π/2

Mathematics - Paper: I - Subjective

Time Allowed: 2:30 Hours | **Maximum Marks:** 80

Note: Note: - Section B is compulsory. Attempt any Three questions from section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Factorize $3x^2 + 3y^2$.

ii. Prove that $\bar{z} = z$ iff z is real.

iii. What is the difference between {a,b} and {{a,b}}?

iv. Show that the statement is a tautology: $\sim q \wedge (p \rightarrow q) \rightarrow \sim p$

v. Define Monoid.

vi. If A = [2 3; 1 5], then show that $4A - 3A = A$.

vii. Find the inverse of matrix [2i i; i -i].

viii. Write any two properties of determinants.

ix. Solve the equation by using quadratic formula: $15x^2 + 2ax - a^2 = 0$.

x. Define a reciprocal equation and give one example.

xi. Prove that $((1+\sqrt{-3})/2)^9 + ((1-\sqrt{-3})/2)^9 = -2$

xii. Discuss the nature of roots of equation $x^2 + 2x + 3 = 0$.

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Define proper rational fraction.

ii. Find a_2 and a_3 of the sequence in which $a_n = na_{n-1}$ and $a_1 = 1$

iii. Which term of A.P; 5, 2, −1, ... is −85.

iv. Sum the series $1/(1+\sqrt{x}) + 1/(1-x) + 1/(1-\sqrt{x}) +$ ... to n terms.

v. Define harmonic mean between a and b. Write its formula also.

vi. Find the sum to n term of series whose nth term is 2n+3.

vii. Write $n(n-1)(n-2)...(n-r+1)$ into factorial form.

viii. Prove that ${}^nC_r = {}^nC_{n-r}$

ix. Calculate number of diagonals of 5 sided figure.

x. Evaluate $(9.9)^9$

xi. Find middle term in the expansion of $(1/x - x^2/2)^{12}$

xii. Expand $(8 - 5x)^{-2/3}$ upto two terms.

Q4.. (Marks 18) Write short answers to any Nine parts. (9 x 2 = 18)

i. Define the word 'Trigonometry'.

ii. Convert 3 radians into degree.

iii. Find sin θ and cos θ when $\theta = -7\pi/4$.

iv. Express $\cos 7\theta - \cos \theta$ as product form.

v. Find the value of sin 2α when $\sin \alpha = 2/3$, where $0 < \alpha < \pi/2$.

vi. Find the value of tan 105°.

vii. Find the period of cos(x/6).

viii. Solve the right triangle, in which $\alpha = 58°13'$, b = 125.7 and $\gamma = 90°$.

ix. Write half angle formulas for sin(α/2) and sin(β/2).

x. By using the cosine and sine law, solve the triangle ABC given that b = 3, c = 5, $\alpha = 120°$

xi. Find the value of $\tan^{-1}(-\sqrt{3})$.

xii. Define trigonometric equation.

xiii. Find the solution of sec x = −2 which lies in the interval [0, 2π].

> Note: SECTION-C (Note: Attempt any Three questions. Each question carries (5+5=10) marks.)

Q5.. (Marks 10) (a) Find the value of x if |3 1 x; 1 3 4; 0 0 1| = −30

(b) Show that the roots of $x^2 + (mx + c)^2 = a^2$ will be equal, if $c^2 = a^2(1 + m^2)$.

Q6.. (Marks 10) (a) Resolve $(x^2 + 1)/(x^3 + 1)$ into partial fraction.

(b) Two dice are thrown twice. What is the probability that sum of dots shown in the first throw is 7 and that of second is 11?

Q7.. (Marks 10) (a) Find 'n' so that $(a^n + b^n)/(a^{n-1} + b^{n-1})$ may be the A.M. between a and b.

(b) Use mathematical induction to prove that x+y is a factor of $x^{2n-1} + y^{2n-1}$, $x \neq -y$.

Q8.. (Marks 10) (a) Prove that $\sin^6\theta - \cos^6\theta = (\sin^2\theta - \cos^2\theta)(1 - \sin^2\theta \cos^2\theta)$.

(b) If $\alpha + \beta + \gamma = 180°$, show that $\cot \alpha \cdot \cot \beta + \cot \beta \cdot \cot \gamma + \cot \gamma \cdot \cot \alpha = 1$

Q9.. (Marks 10) (a) Solve the triangle ABC in which $a = \sqrt{3} - 1$, $b = \sqrt{3} + 1$ and $\gamma = 60°$.

(b) Prove that $\sin^{-1}(77/85) - \sin^{-1}(3/5) = \cos^{-1}(15/17)$